

Outcome-Based Reuse of Cellular Molecular Receivers: Response Prediction and Calibrated Waiting

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Abstract—This paper studies when a living cell used as a molecular receiver can be reused, since its response to a new command depends on a hidden biochemical memory of earlier commands. We argue that reuse must be judged by the outcome of the next command rather than by average response-prediction error, and test this with 1,565 measured FGF2–ERK single-cell trajectories and a published mechanistic signaling model (B3). First, on measured cells, reporter filtering plus a mechanistic increment lowers the 10-min forecast error by 6.7–7.6% against persistence, but in the windows before a new command the increment raised the error. Second, in B3 simulations with disjoint reference, calibration and test receivers and a 90% success requirement certified for each previous command, feedback on a smoothed reporter or the model’s belief finishes 6.6–7.5 min earlier than certified fixed waits (medians over 20 splits) at about 95% test success; across six predictors, including ARX and NARX baselines, the 10-min forecast error ranks these decisions poorly (median rank correlation 0.17). Third, this gain is paid in slack, the time from readiness to the probe: feedback leaves about half the median slack of fixed waits, so a recovery error hidden from the reporter, like the one measured probes reveal in B3, costs it reliability first; a 10% slower recovery broke the requirement of every rule calibrated on the model, and one probe per calibration receiver at randomized parameters rarely sufficed to certify a threshold. Reusing a cellular receiver therefore needs measured task outcomes, not reporter forecasts alone.

Index Terms—Molecular communication, cellular receivers, mechanism-informed prediction, Bayesian filtering, receiver reuse, waiting rules, FGF2, ERK.

I. INTRODUCTION

Molecular communication (MC) with living cells as receivers offers a natural interface between engineered communication systems and biological processes. Unlike a conventional receiver, a cell retains the effect of earlier chemical commands in its receptors and signaling feedback, so a receiver that has just responded may not execute the next command reliably. This raises a communication-level question for repeated molecular tasks: *when can a living receiver be used again, and what information does that decision need?*

Receiver memory is familiar in MC, but it is attributed to the channel. Residual molecules cause intersymbol interference (ISI) [1], [2], and receivers counter it with sequence detection [3], memory-aware decoders [4], adaptive thresholds [5] and symbol or detection intervals that trade rate against residual ISI [6], [7]; reception itself is modeled as ligand binding or absorption [8], and the guard interval is set in advance.

Living receivers are now being characterized and engineered, from yeast pheromone signaling [9] and long-term closed-loop experiments [10] to astrocyte calcium signaling [11], genetic-circuit design [12] and error control in cell signaling [13], and learning-based designs train MC transceivers and channel models from data [14]–[16]. Systems biology shows that a cell’s signaling response depends on the history and frequency of earlier stimuli [17], [18], and feedback control of single cells is feasible [19], [20]. In a cellular receiver this memory is a hidden biochemical state seen only through a noisy reporter, so deciding when to send the next command is a stopping problem, related to sequential detection [21], [22] and optimal stopping [23].

To our knowledge, this decision has not been studied on measured cellular receivers. Prediction studies report average forecast errors, but a receiver is reused when the next command succeeds, so three points remain open: whether information that improves average prediction also helps where the next command arrives; whether reporter feedback can replace a fixed guard interval when a reliability requirement must hold after every previous command, evaluated on receivers that did not shape the rule; and what such a gain costs when calibration outcomes are scarce or the model misjudges recovery.

We combine experimental FGF2–ERK single-cell trajectories with the published B3 signaling model [17], executed without refitting. Nested predictors add reporter filtering, a mechanistic B3 increment and explicit input history one at a time. Reuse is posed as a waiting rule with a success requirement for each previous command, simulated on B3 with disjoint reference, calibration and test receivers and certified by binomial tests; the same simulation links predictors to decisions, limits calibration to one probe outcome per receiver, and varies the task and the form of model error. Measured probe pulses test B3’s recovery. Details are in the supplementary material (Supplements S1–S13). Our contributions are:

- *Information value on measured cells.* Filtering plus the B3 increment beats persistence on all four evaluation protocols, the increment carrying the gain, whereas explicit input history adds little; in the next-command window at 25 ng/ml the increment raised the 10-min error, consistent with B3 overestimating recovery after a long, strong command.
- *Reuse decisions under explicit simulation conditions.* On

held-out simulated receivers, feedback on a smoothed reporter or the B3 belief finishes 6.6–7.5 min earlier (medians over 20 splits) than certified history-conditioned fixed waits at the same per-history success, reported with confidence bounds; across six predictors, the 10-min forecast error ranks decision quality poorly, whereas the error of the predicted task outcome ranks it better.

- *A speed–robustness trade-off.* Feedback gains its time by spending slack, the time from readiness to the probe: after the 30-min command it left a median slack of 8–10 min against 20 min for fixed waits, and under a recovery delay hidden from the reporter a rule’s success equals the share of receivers whose slack covers the delay. Feedback therefore loses reliability first under recovery errors far smaller than the one a measured probe suggests, and one probe per calibration receiver at randomized parameters rarely supplied the outcome calibration that could restore it.

II. SYSTEM AND TASK MODEL

Let $u(t) \geq 0$ be the commanded FGF2 concentration (ng/ml). Delivery to the receiver is represented abstractly as

$$\begin{aligned} \tilde{u}(t) &= \mathcal{T}_\theta[u](t), \quad \dot{x}_j = f_{\theta_j}(x_j, \tilde{u}), \\ z_j &= g(x_j), \quad y_{j,i} = z_j(t_i) + \nu_{j,i}, \end{aligned} \quad (1)$$

where $\tilde{u}$ is the local concentration, x_j the hidden state of cell j , z_j its noise-free normalized ERK reporter and $y_{j,i}$ the observation at frame t_i with noise $\nu_{j,i}$. Neither $\mathcal{T}_\theta$ nor $\tilde{u}$ is calibrated here. The information $\mathcal{H}_t$ holds the executed commands and the observations that have arrived by t ; Fig. 1 sketches the setting.

Reuse task. The amplitude and duration of the next command are fixed; only its timing is decided. A previous command of $m \in \{3, 30\}$ min (the history) ends at time 0, and a 5-min probe (the next command) at the same amplitude acts on the receiver at $\tau \in \{0, 2, \dots, 120\}$ min. With z_j^τ the response when the probe acts at τ and r_j^{nv} the naive rise, the same receiver’s rise to the probe without a previous command,

$$\begin{aligned} S_j(\tau) &= \mathbf{1} \left\{ \max_{s \in [\tau, \tau+W]} z_j^\tau(s) - z_j^\tau(\tau) \geq \kappa r_j^{\text{nv}} \right\}, \\ R_{j,t} &= \mathbb{P}(S_j(t) = 1 \mid \mathcal{H}_t), \end{aligned} \quad (2)$$

with window $W = 20$ min and $\kappa = 0.5$. This is a proxy task: it asks that the receiver respond again with half its naive amplitude, not that it perform a biological function; $\kappa = 0.8$ is infeasible, and Sec. V-G varies κ , W and the failure penalty. Success is scored on the noise-free response; decisions see only noisy observations. The completion time is $T = \tau + W$ on success and $T_f = 140$ min, the latest completion of any success, on failure. The design problem is

$$\min \mathbb{E}[T \mid m] \quad \text{s.t.} \quad \mathbb{P}(S = 1 \mid m) \geq q, \quad m \in \{3, 30\}, \quad (3)$$

with $q = 0.9$. The probabilities refer to a stated population of receivers: in the simulations, a receiver is a distinct B3

posterior parameter vector with an independent draw of reporter noise, so they express uncertainty about the population-mean parameters of one model, not cell-to-cell heterogeneity. Requiring q for each history, not on average, prevents a rule from buying speed after one history with failures after the other. An oracle probes at the first decision time at which the probe would succeed. Two delays can enter a rule: a feedback delay d , when the reporter reaches the controller late so that a probe decided at t acts at $t + d$, and a post-trigger delay δ , an extra wait after the readiness estimate crosses its threshold (Sec. III-C).

III. PREDICTORS AND WAITING RULES

A. Nested Predictors

Predictors for horizons $h \in \{2, 10, 20, 30\}$ min are fitted by ridge regression with equal weight per condition and add information one step at a time: **Persist** ($\hat{y}_{t+h} = y_t$); **Current** (current command, reporter, slope and the known future command); **Filter** (plus causal reporter filters); **Lag** (lagged values instead of filters); **Filter+B3** and **Lag+B3** (plus a B3 increment); and **F+B3+Input** (plus input filters and cumulative exposure; Table S1). The causal filters with scales $\ell \in \{2, 10, 60\}$ min are

$$\begin{aligned} \bar{y}_{\ell,i} &= e^{-\Delta_i/\ell} \bar{y}_{\ell,i-1} + (1 - e^{-\Delta_i/\ell}) y_i, \\ v_\ell(t) &= \frac{1}{\ell} \int_0^t e^{-(t-s)/\ell} u(s) ds, \end{aligned} \quad (4)$$

with $\bar{y}_{\ell,0} = y_0$ and $\Delta_i = t_i - t_{i-1}$. As mechanism we execute the authors’ B3 model with their posterior samples. With $f_s(t)$ the noise-free reporter of sample s under the recorded command, the samples are weighted by the cell’s past reporter,

$$\begin{aligned} w_s(t) &\propto \exp \left[- \sum_{t_i \leq t} (y_i - f_s(t_i))^2 / (2\sigma^2) \right], \\ \hat{\delta}_{t,h} &= \sum_s w_s(t) [f_s(t+h) - f_s(t)], \end{aligned} \quad (5)$$

a fixed-parameter particle approximation of the B3 belief that uses the same information as the learned predictors; since $\hat{\delta}_{t,h}$ already uses the command history, **F+B3+Input** tests only the extra value of explicit input features.

B. Waiting Rules and Certified Calibration

A waiting rule probes at the first decision time at which its readiness estimate reaches a threshold c , or at 120 min,

$$\tau = \min (\{t : \hat{R}_t \geq c\} \cup \{120\}), \quad (6)$$

a stopping time on the observations, as in sequential tests [21]; we compare members of this threshold family and claim no optimality beyond it. Three estimators of R_t are compared: *current* uses only y_t , through a Gaussian kernel over the (reporter, success) states of reference receivers; *smoothed* applies the kernel to a causal exponential smoothing of the observations; the *B3 belief* weights reference receivers by all observations as in Eq. (5). The comparator is one fixed wait per history, a history-conditioned guard interval.

Roles. The simulated receivers are the 948 distinct vectors of the B3 posterior file (which repeats 52 of its 1,000 rows).

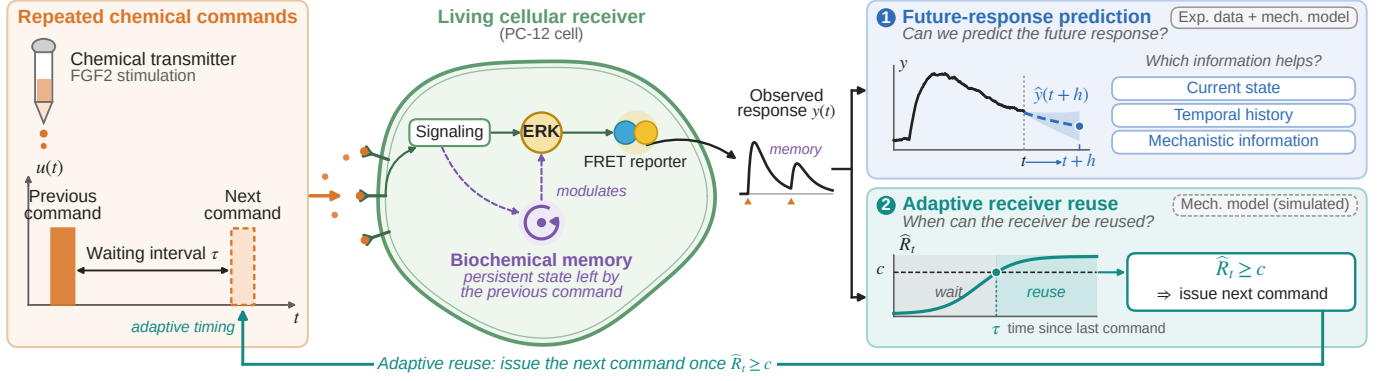


Fig. 1. System concept. Repeated FGF2 commands drive a living receiver whose biochemical memory shapes the ERK/FRET response $y(t)$ to later commands. The response supports (1) prediction of $\hat{y}(t+h)$ and (2) readiness-based reuse, where the next command is sent once $\hat{R}_t \geq c$.

Each of 20 random splits divides them into reference, calibration and test receivers (316 each): references define the estimators, parameters are chosen on calibration receivers, and test receivers are scored once. Split 0 is primary; the others measure stability of one finite posterior set, not independent experiments.

Certification. For each rule and history m the parameters are ordered from conservative to aggressive (thresholds 0.99 down to 0.50, waits 120 down to 0 min, one chain per smoothing scale), $\theta_1, \theta_2, \dots$, and admitted while a one-sided binomial test rejects $\mathbb{P}(S = 1 \mid m, \theta) \leq q$,

$$\Theta_m = \{\theta_k : \mathbb{P}(\text{Bin}(n, q) \geq N_m(\theta_j)) \leq \alpha \quad \forall j \leq k\}, \quad (7)$$

where $N_m(\theta)$ counts the $n = 316$ calibration receivers that succeed under θ and $\alpha = 0.025$ (split over the smoothing chains). This fixed-sequence procedure controls the probability of admitting any parameter whose success is at or below q [24]; with 316 receivers it needs at least 295 successes (93.4%). The fastest admitted parameter on calibration is used. For receivers exchangeable with the calibration receivers, the probability that an admitted parameter has success at or below q for either history is then at most 0.05; this concerns calibration, and test success is reported separately. If Θ_m is empty, the rule probes at 120 min after history m , and these receivers stay in all statistics. Plug-in targets (smallest calibration completion with calibration success ≥ 0.90 or 0.94) serve as secondary arms; test success is reported with one-sided Clopper–Pearson bounds [25].

C. From Prediction to Decision and from Few Probes

To test whether forecast accuracy carries over to decisions, six predictors estimate the recovery statistic $\rho(t)$, the probe rise within W over the naive rise, so that $S = 1\{\rho \geq \kappa\}$, from $\mathcal{H}_t$: the history mean of $\rho(t)$ (the information of a fixed wait), the current and smoothed kernels, ARX and NARX (ridge regressions on the last five observations, time and history, NARX with all quadratic terms) and the B3 belief. A monotone map fitted on out-of-fold predictions of the reference receivers turns $\hat{\rho}$ into a success probability,

which Eq. (6) thresholds with a certified c . Each predictor also forecasts the reporter 10 min ahead; for the two kernels this forecast is the current or smoothed value itself. The predictors estimate ρ directly; only the B3 belief derives it from a response model. To limit calibration information, each calibration receiver is instead probed once, at its decision time under a randomly assigned rung of a 20-rung parameter ladder, for 50–200 receivers per history. The estimators keep their 316 reference receivers with complete simulated curves, which cost no probes, and the ladders, the smoothing scale and the delay trigger were set from the full-information choices, so the budgets count only the selection of a threshold or wait. A plug-in selection fits a monotone success curve over the rungs; a certified selection applies Eq. (7), counting for rung j the receivers probed at rung j or a more aggressive one, whose outcomes cannot exceed their outcome under rung j , so the test stays conservative (Supplements S10, S11).

IV. DATA AND EVALUATION

All files come from the paper-linked repository at a pinned commit [26], [27]: 22 conditions of PC-12 cells (1,565 cells, 2-min frames; Table S2) with roles frozen before any newly used response file was downloaded. Two protocols train (sustained, 3/20 pulses), the single 5-min pulse validates, and three test: A, a new single 10-min pulse; B, a single 60-min pulse with inferred timing; and C, the mixed protocol at the new concentrations 0.25 and 25 ng/ml, recorded with its earlier-inspected blocks at 2.5 and 250 ng/ml, which enter only Sec. V-F. Every commanded schedule was traced to the authors' files: the mixed protocol's recorded pulses at 43–46, 66–96 and 156–161 min, cut at 42 min by the authors' normalization, give the model input 1–4, 24–54 and 114–119 min (a 3-min, a 30-min and a 5-min pulse), and all 23 condition schedules agree with the authors' model files (Supplement S8). The authors fitted B3 to the sustained and 3/20 means and compared architectures on the 5-min and mixed blocks, all at 2.5 and 250 ng/ml, so B3 is a calibrated reference, not a blind model. The score is the root of the condition-equally weighted mean squared error (Eq. (S1)); the noise-adjusted skill is $1 - (\text{MSE} - \hat{\sigma}_\nu^2) / (\text{MSE}_{\text{Persist}} - \hat{\sigma}_\nu^2)$

TABLE I
10-MIN FORECAST RMSE ON MEASURED CELLS.

Model	Val.	A	B	C	A, 2 min
Persist	0.02353	0.02287	0.02651	0.02876	0.01302
Current	0.02503	0.02442	0.02715	0.02825	0.01342
Filter	0.02403	0.02220	0.02715	0.02827	0.01283
Lag	0.02343	0.02272	0.02565	0.02769	0.01300
Filter+B3	0.02197	0.02120	0.02448	0.02658	0.01263
Lag+B3	0.02161	0.02114	0.02474	0.02652	0.01270
F+B3+Input	0.02195	0.02098	0.02439	0.02656	0.01260
Filter < Current	4/4	4/4	2/3	1/2	4/4
Filter+B3 < Filter	4/4	3/4	2/3	2/2	3/4
Lag+B3 < Lag	4/4	4/4	3/3	1/2	4/4
F+B3+Input < Filter+B3	1/4	2/4	2/3	1/2	2/4

with $\hat{\sigma}_\nu$ a robust estimate of white reporter noise (0.006–0.009). Forecast origins are assigned to phases: stimulation, next command (a pulse starts in $(t, t + h]$), early washout (≤ 30 min after a pulse) and late. Cells share a session and no day identifiers exist, so cell-bootstrap intervals describe cell sampling, not independent experiments. Measured probes differ from the simulated task in probe length, naive reference and noise, so they check B3 but do not evaluate Eq. (2); only the simulations do (Table S5). Every analysis was frozen in its own protocol before running, but all are post hoc with respect to inspected data.

V. RESULTS AND DISCUSSION

A. Which Information Predicts the Response

Table I reports the 10-min RMSE for the validation (Val.) and test roles (last column: 2 min on Test A); bold marks the best predictor per column, and the bottom rows count the conditions in which the first predictor of each pair has the lower MSE. Filter+B3 beat persistence on all four roles, by 6.7–7.6%, and the B3 increment carried this gain: it lowered the error of Filter on every role (4.5–9.8%) and in five of six held-out protocols of a leave-one-protocol-out analysis with tuning on the five remaining protocols only, whereas at 10 min filtering helped only on the validation protocol and Test A. On Test A, filtering, the B3 increment and input history reduced the error by 9.1%, 4.5% and 1.0% relative to their predecessors (cell-bootstrap intervals 7.4–10.9%, 3.0–6.1% and 0.6–1.4%). At 20–30 min the increment raised the noise-adjusted skill on Test A to 21% against 9–13% for Filter, whereas explicit input history destabilized longer forecasts (Supplement S2).

B. The Next-Command Window

By phase, the B3 increment reduced the error of Filter during stimulation, early washout and the late phase (by 4.6–8.9%) but raised it in the next-command window from 0.03306 to 0.03544, only at the 10-min horizon. That window holds the two mixed test conditions: the increment helped at 0.25 ng/ml and hurt at 25 ng/ml (0.02920 to 0.03603), and phase-gated variants fitted on the training protocols, which contain no probe after a long command, did not repair this. An average gain in prediction thus need not hold where the next command

TABLE II
CERTIFIED WAITING ON HELD-OUT SIMULATED RECEIVERS (B3, NOISE S.D. 0.005). S_m : TEST SUCCESS (%) AFTER THE m -MIN COMMAND, ONE-SIDED 95% LOWER BOUND IN PARENTHESES; T : MEAN COMPLETION (MIN); ΔT : PAIRED DIFFERENCE TO THE FIXED WAITS WITH 95% BOOTSTRAP INTERVAL (PRIMARY SPLIT); MET: SPLITS (OF 20) WITH $S_3, S_{30} \geq 90\%$, IN PARENTHESES THOSE CERTIFIED FOR BOTH HISTORIES.

Rule	S_3 (LB)	S_{30} (LB)	T	ΔT [95% CI]	Met
Fixed (94/108)	94.9 (92.4)	94.0 (91.3)	122.0	–	20 (20)
Current	94.9 (92.4)	94.6 (92.0)	122.1	+0.1 [–1.1, +1.3]	20 (20)
Smoothed	94.6 (92.0)	94.6 (92.0)	114.2	–7.8 [–8.8, –6.8]	20 (20)
B3 belief	94.6 (92.0)	94.6 (92.0)	114.7	–7.3 [–8.3, –6.3]	18 (16)
Oracle	100.0 (99.1)	98.4 (96.7)	104.8	–	–
Noise s.d. 0.01 (same roles; the fixed waits do not change)					
Current	94.6 (92.0)	93.7 (90.9)	129.8	+7.7 [+6.7, +8.7]	20 (20)
Smoothed	94.6 (92.0)	94.6 (92.0)	117.5	–4.5 [–5.5, –3.5]	20 (20)
B3 belief	93.4 (90.6)	93.4 (90.6)	115.0	–7.0 [–7.8, –6.2]	19 (20)

arrives (Supplement S3); Sec. V-F examines B3’s recovery directly.

C. Waiting under a Certified Per-History Requirement

No recorded protocol varies the wait, so we simulate waiting on B3, which carried the predictive gain; Secs. V-F and V-G test where this is unsafe. With 25 ng/ml commands and observations every 2 min, waits of 6–48 min never succeeded. On the primary split (Table II) every rule was certified for both histories and met q on the 316 test receivers, at 94.0–94.9% success with one-sided 95% bounds of 91.3–92.4%. Feedback on a smoothed reporter or the B3 belief finished 7.8 and 7.3 min earlier than the certified fixed waits (94 and 108 min), with paired intervals excluding zero; over 20 splits (medians 7.5 and 6.6 min) smoothing was faster in all and met q in all, the B3 belief was faster in 16 and met q in 18. In four splits its belief, overconfident at the top, failed certification at the first threshold after the 3-min command, so it fell back to the 120-min probe there. The current reporter alone gained nothing. A plug-in target equal to the requirement (0.90) met q in only 10–14 of 20 splits; the certified margin, derived from the calibration sample size, was no slower than the target 0.94 that our earlier evaluation had chosen after seeing its results. With 316 test receivers a success of 94–95% has a 97.5% lower bound near 91–92%, so the test data alone confirmed q for both histories at joint level 0.05 in only 11–17 of 20 splits: the evidence supports an empirical success rate with a confidence bound, not a guarantee. At noise s.d. 0.01 smoothing gained 4.5 and the B3 belief 7.0 min on the primary split (medians 3.9 and 5.9 min), whereas the current reporter was 7.7 min slower than the fixed waits (median). The repeated posterior vectors had given some earlier evaluation receivers an identical twin among the references; re-scoring that evaluation without them, with its rules unchanged, moved its gains by less than 0.1 min (Supplement S8). With a feedback delay the rule must predict readiness; in the earlier evaluation a B3-predictive rule kept an advantage over a smoothed trend when the reporter was

TABLE III

FROM PREDICTION TO DECISION (B3 SIMULATION, MEDIAN OVER 20 SPLITS): 10-MIN REPORTER FORECAST ERROR e_{10} AND ERROR e_p OF THE RECOVERY STATISTIC AT 60–120 MIN ($\times 10^{-3}$), BRIER SCORE ($\times 10^{-3}$), FALSE-READY RATE AFTER THE 3/30-MIN COMMAND, COMPLETION T AND ΔT AGAINST CERTIFIED FIXED WAITS (MIN).

Predictor	e_{10}	e_p	Brier	FR (%)	T	ΔT	Met
Hist. mean	3.7	49	45	6.0/6.0	121.9	+0.0	20/20
Current	11.8	64	65	4.3/4.5	122.8	+1.3	20/20
Smoothed	14.5	27	25	4.3/4.4	114.4	-7.5	20/20
ARX	4.8	38	34	5.2/6.6	117.1	-4.6	20/20
NARX	3.2	37	30	4.9/4.2	115.9	-5.7	20/20
B3 belief	2.3	27	26	4.4/4.4	114.1	-7.7	19/20

noisy or sparsely sampled, provided B3 is right (Supplement S5).

D. Forecast Error versus Decision Quality

Table III follows six predictors from forecast to decision on the same test receivers. The B3 belief, correct by construction in this simulation, had the smallest forecast error and the shortest completion. Otherwise forecast error and decisions separated: the smoothed kernel had the largest forecast error, because its forecast is by definition the lagging smoothed value, yet the smallest e_p and Brier score and a gain of 7.5 min; the history mean forecast the reporter better than ARX and the kernels but, holding only the information of a fixed wait, gained nothing; NARX had the second-smallest forecast error but gained 2 min less than the B3 belief. Over the six predictors the pre-registered rank correlation with completion had a median over splits of 0.17 for e_{10} and 0.40 for the error of $\hat{p}$ over all decision times, and on the primary split the orderings by e_p at 60–120 min and by the Brier score matched the ordering by completion except for the first two places. Every predictor met q in 19–20 of 20 splits, with 4–7% false ready calls. Even among the four predictors that forecast the reporter with a model, the history mean was third in forecast error and gained nothing. Reporter forecast accuracy is therefore a weak proxy for reuse; what matters is the accuracy of the predicted task outcome near the decision.

E. Calibration from One Probe per Receiver

The calibrations above use every calibration receiver's success at every decision time, information no experiment provides: a cell gives one probe outcome after one history. Fig. 2 limits calibration to one probe per receiver (Sec. III-C). Plug-in calibration from one probe kept the time gain of smoothing (6.0–7.2 min against equally calibrated fixed waits) but met q for both histories in only 1–9 of 20 splits at 50–200 receivers per history. Certified calibration from one probe admitted no fixed wait and no current-reporter threshold in any split, and a smoothing or B3-belief threshold for at least one history in at most 8 of 20 splits (for both histories in at most one), so these rules mostly fell back to the 120-min probe and their median completion stayed at 140 min: with uniformly assigned rungs most probes test aggressive parameters, which keeps the conservative count below the admission level. Only

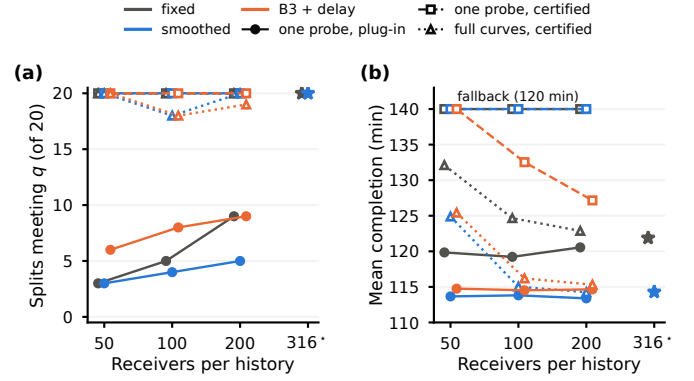


Fig. 2. Calibration from one probe per calibration receiver (B3 simulation, model correct): (a) splits meeting q for both histories on test, (b) median completion over 20 splits, for fixed waits, smoothing and the B3 belief with a post-trigger delay, by calibration information and selection. One-probe certified rules reach 20 of 20 in (a) mostly by falling back to the 120-min probe. Stars: certification on complete outcome curves of 316 receivers (Table II).

the narrow ladder of a post-trigger delay after the B3 belief crossed 0.90 was certified at 100 and 200 receivers, finishing at 132.5 and 127.2 min, later than fixed waits certified on complete outcome curves (121.9 min). Complete curves of the same 200 receivers certified fixed waits and smoothing (122.9 and 114.2 min). The full-curve results are thus an ideal-information benchmark; at these budgets and with randomized rungs, one-probe calibration supplied either the speed or the certified reliability, not both; probe designs that concentrate or sequence the probes may need fewer receivers and remain open (Supplement S10).

F. Where B3 Errs: Natural Probes in Measured Cells

Two recorded protocols contain later pulses that act as probes at fixed waits (25 ng/ml), well before the certified waits: the second 3-min pulse of the 3/20 protocol, 20 min after the first (58 cells; naive rise: the cell's own first pulse), and the 5-min pulse 60 min after the 30-min pulse of the mixed protocol (84 cells; naive rise: median of the single 5-min pulse cells). B3 predicted no success at either probe and every estimator called no cell ready, so the rules made no false ready call but missed the 19% of cells that met the criterion at the first 3/20 probe; none did at the mixed probe. A post hoc posterior predictive check across doses (Fig. 3(a)), frozen after an exploratory look at the same quantities, shows that 60 min after a 30-min pulse the measured rise at 25 and 250 ng/ml was 4% and 8% of the naive rise, indistinguishable from no response and below all 1,000 posterior samples (5–95% ranges 23–39% and 27–40%), whereas at 2.5 ng/ml it lay within B3's range; after a 3-min pulse it lay within or at the edge of B3's range at the fitted doses and above it at 25 ng/ml. B3 thus errs where its fit gave it no data, recovery after a long, strong command, and no reweighting of its samples can repair this. On measured cells the B3 weights collapsed (effective sample size 1.0–1.4), as on the cells B3 was fitted to (Supplement

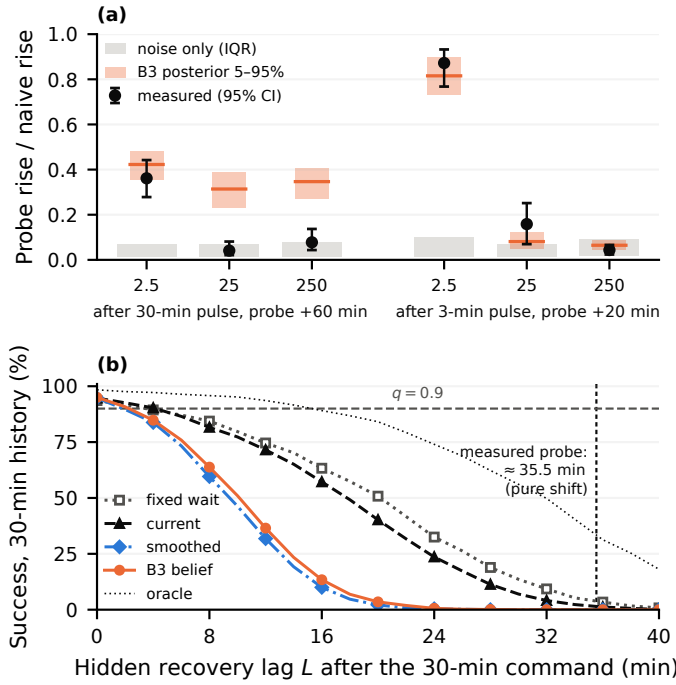


Fig. 3. B3's recovery error on measured cells and its cost for waiting. (a) Probe rise over naive rise at 2.5, 25 and 250 ng/ml after a 30-min pulse (probe 60 min later) and a 3-min pulse (20 min later): measured median with 95% cell-bootstrap interval, B3 posterior predictive 5-95% range (bar; line: median) and the interquartile range expected from noise alone (gray). (b) B3 simulation (earlier calibration: target 0.94, leave-one-out references; noise s.d. 0.005): median success after the 30-min command over 20 splits versus a recovery lag L hidden from the reporter, which equals the share of receivers whose slack is at least L ; dashed line: requirement; dotted vertical line: the measured mixed probe read as a pure time shift.

S6); a B3-belief rule on real cells would need process noise or cell heterogeneity.

G. The Cost of Speed: Slack and Model Error

Every rule above, fixed waits included, is calibrated on B3 outcomes, so none is model-free, and Sec. V-F shows that B3 overestimates recovery after a long command. Let the slack of a receiver be the time from its readiness onset to the probe. Because readiness is monotone in time, a probe still succeeds under a recovery delay L hidden from the reporter exactly when its slack is at least L , so a rule's success under such a delay is the share of receivers with slack $\geq L$: robustness to a hidden recovery error is the slack distribution. Calibrated to the same success, the rules keep a similar lower tail after the 30-min command (10% quantile 1-4 min, set by the requirement) but place the rest differently: the fixed waits left a median slack of 20 min, the current rule, which gained nothing, 16 min, and the smoothed and B3-belief rules only 8 and 10 min (earlier calibration, medians over 20 splits). Feedback thus buys its speed with slack. Fig. 3(b) shows the cost: because the lower tail is thin for every rule, the fixed waits and the current rule kept the requirement only up to a 2-min lag and the feedback rules lost it at 2 min, but beyond that feedback fell much faster (at an 8-min lag, 60-64% success against 84% for the fixed

waits). The strict evaluation confirms this and extends it to other error forms, all hidden from the reporter (Supplement S12): with rules certified on the model, a 4-min shift left the requirement only to the fixed waits and the current rule (10 and 6 of 20 splits), a 10% amplitude loss or a 10% slower recovery broke it for every rule in every split, and a spread of recovery speeds across receivers (log-normal, $\sigma = 0.1$) for all but the fixed waits, which met it in 2 of 20; at a 20% amplitude loss or a 25% slowing even the oracle missed q within 120 min. Recertification on complete outcome curves restored q in nearly every split in which the oracle met it, but the threshold rules then mostly fell back after the 30-min command; only at the 4-min shift did smoothing keep a gain (7.0 min), and under the speed spread re-certified smoothing and the B3 belief were 2.4 and 13.1 min slower than the fixed waits; a post-trigger delay kept 7.5 min at an 8-min lag with complete curves of 200 receivers (Supplement S10). The conclusions did not depend on the task details: the gain held at $\kappa = 0.4$ (5.0-5.4 min) and at larger failure penalties (7.2-9.4 min at 260 min); the window W did not matter, because the probe response reaches the success level within 10 min, and at $\kappa \geq 0.6$ the task was infeasible after the 30-min command.

How large is the measured error? In the simulated task (a 30-min command alone; the mixed protocol also has an earlier 3-min pulse), B3's posterior-median probe rise reaches the measured 4% of the naive rise at a wait of 24.5 instead of 60 min, which under a pure time shift is a B3-equivalent lag of about 35.5 min (Fig. 3(b)). One probe cannot tell a delay from a loss of amplitude, a slower recovery or a mixture of responders, and the measured rise is at noise level, so this is an equivalent value, not a bound. Read instead as an amplitude loss (to 12.5% of B3's rise) or a slowing (by 2.45 times) fitted to the same point, it leaves no receiver ready within 120 min after the 30-min command.

H. Implications and Evidence Boundaries

A previous command acts on the next one like ISI whose memory sits in the receiver's biochemistry [2], [6], and its length depends on that command. A fixed wait is a guard interval, conditioning it on the previous command is the first step, and feedback adapts to state variability within each history, much as adaptive detection adapts to residual interference [4], [5]. A guard interval buys robustness with slack, and feedback sells that slack for speed; its reliability therefore rests on calibrating recovery timing against true outcomes, and the experiments that supply them are expensive: with one probe per cell at randomized parameters, up to 200 cells per history rarely certified a threshold here, and only a slow post-trigger delay rule was certified for both histories. A prospective test should randomize 3- versus 30-min preconditioning per chamber, place probes near and after the rule's trigger at several doses, measure the naive response in same-session controls, and count sessions rather than cells. *Boundaries.* The prediction comparisons are post hoc on inspected roles from one session; the waiting results come from one simulated model structure, whose posterior expresses parameter uncer-

tainty, not cell heterogeneity, and whose vectors were seen while the rules were developed, so the 20 splits cover role sampling and reporter noise only; full-curve calibration is an ideal-information benchmark; the mismatch forms are stress tests, and only the anchored readings use a measurement. No waiting rule has been run on cells, so these results do not show that feedback-driven reuse works in real cells.

VI. CONCLUSION

On experimental FGF2–ERK trajectories, filtering plus a calibrated mechanistic increment predicted the response better than persistence, although the increment raised the error before a new command, and measured probes showed B3 overestimating recovery after a long, strong command. In B3 simulation with disjoint roles and per-history reliability certified on calibration receivers, feedback on a smoothed reporter or the B3 belief finished 6.6–7.5 min earlier (medians over 20 splits) than history-conditioned fixed waits on held-out receivers, and a predictor’s value for this decision followed its accuracy on the task outcome, not its average forecast error. This gain is paid in slack: feedback probes closer to readiness than a guard interval, so recovery errors hidden from the reporter, smaller than the one a measured probe suggests, cost it reliability first, and the outcome calibration that could restore it was rarely certifiable from one probe per calibration receiver at randomized parameters, at up to 200 receivers per history. Reusing a cellular receiver therefore requires calibration on measured task outcomes, and whether feedback helps real cells must be tested by running the selected rules on cells.

DATA AND CODE AVAILABILITY

Code, frozen results, protocols with hash records, the supplementary material and verification scripts are available at <https://github.com/ruifeng105/task-aware-cellular-mc>; one command re-runs the analysis and checks the reported values against a frozen reference. Sources are credited in [17], [26], [27]; author files keep their attribution and receive no new license.

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